Agenda

- Last time
  - PRNG
  - Uniform distribution
  - Random subset
- Today

Hello, "Random" World!

```
#include <iostream>
#include <random>
int main() {
    std::mt19937 mt(1729);
    std::uniform_int_distribution<int> dist(0, 99);
    for (int i = 0; i < 16; ++i) {
        std::cout << dist(mt) << " ";
    }
    std::cout << std::endl;
}
```

Engine: $[0, 2^{32})$

Deterministic 32-bit seed

Distribution: [0, 99]

Note: [inclusive, inclusive]

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Last Time

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  - PRNG
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  - Random subset
- Example
  - Two midsemester question papers
  - Want to “randomly” assign papers

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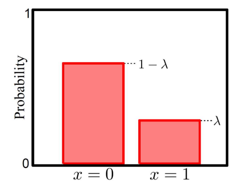
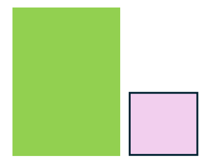
Last Time

- Last time
  - PRNG, Uniform distribution
  - Random subset
- Example
  - Two midsemester question papers, want to randomly assign papers
  - Room size constraint, one room has only ~30% space





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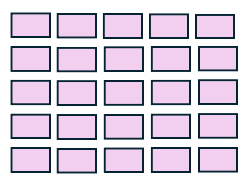
Last Time

- Last time
  - PRNG, Uniform, Random subset
- Example
  - Three different TAs with different capacity to grade
  - 0.2, 0.5, 0.3



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Generating a discrete pmf

- Generate a discrete random variable X having the pmf
  - Example: X = 0, 1 and 2 and 3 with probabilities 0.2, .15, .25, 0.4
- Algorithm
  - Generate a sequence of random numbers U that is uniformly distributed over [0, 1]

$$P\{X = x_j\} = p_j, \quad j = 0, 1, \cdots, \sum_j p_j = 1$$

$$X = \begin{cases} x_0 & \text{If } U < p_0 \\ x_1 & \text{If } p_0 \leq U < p_0 + p_1 \\ \vdots & \vdots \\ x_j & \text{If } \sum_{i=0}^{j-1} p_i \leq U < \sum_{i=0}^{j} p_i \\ \vdots & \end{cases}$$

$$0 < a < b < 1 \longrightarrow P\{a \leq U < b\} = b - a$$

$$P\{X = x_j\} = P\left\{\sum_{i=0}^{j-1} p_i \leq U < \sum_{i=0}^{j} p_i\right\} = p_j$$

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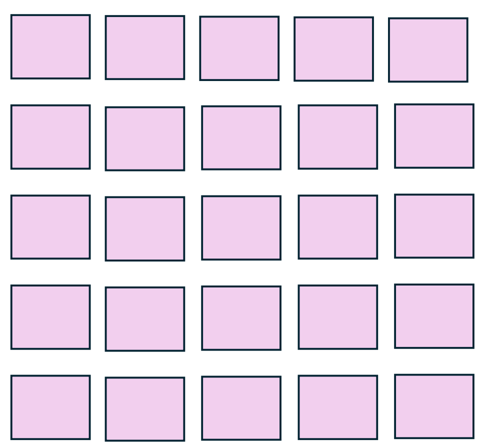
Last Time

- Last time
  - PRNG, Uniform distribution
  - Random subset: Any of the $\binom{n}{k}$ possibilities should be equally likely
- Example
  - Moderation: Choose k out of n
  - 2 out of 5



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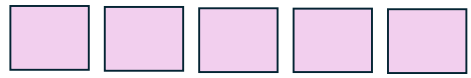
$$I_j = \mathbf{1}\left\{U_j < \frac{k - I_1 - \cdots - I_{j-1}}{n - j + 1}\right\}$$

k=2, n=5

Example

- Samples
  - 0.79 0.48 0.19 0.24 0.97
    1. RHS = 2/5 Do not select Item 1 since $0.79 \nless 0.4$
    2. RHS = 2/4 = 0.5. Select item 2 since 0.48 < 0.5
    3. RHS = 1/3 = 0.33. Select 3 since 0.19 < 0.3
    4. We are done, we have selected 2 items



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Last Time

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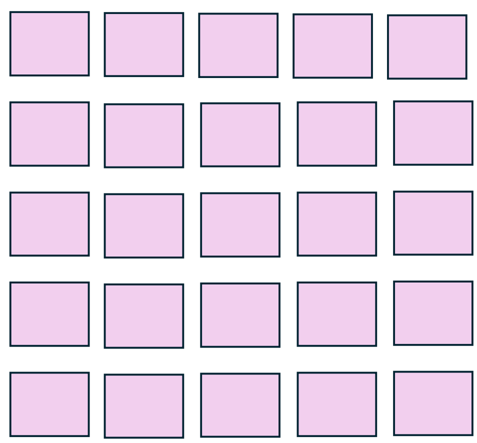
- Example
  - Moderation
  - 2 out of 5



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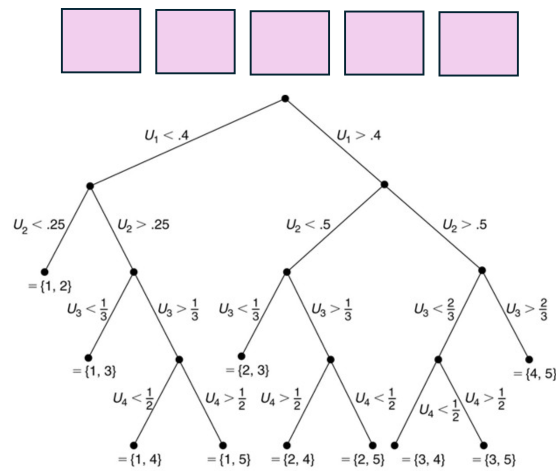
$$I_j = \mathbf{1}\left\{U_j < \frac{k - I_1 - \cdots - I_{j-1}}{n-j+1}\right\}$$

k=2, n=5

Example

- Samples
  - 0.79 0.48 0.19 0.24 0.97
- Chance of getting to any of the leaf nodes is exactly any of the $\binom{n}{k}$



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Agenda

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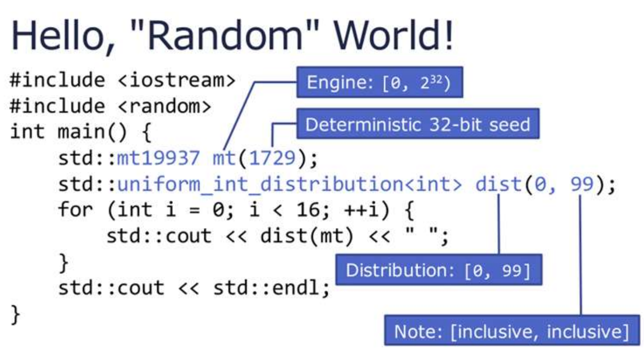
- Today
  - Normal distribution



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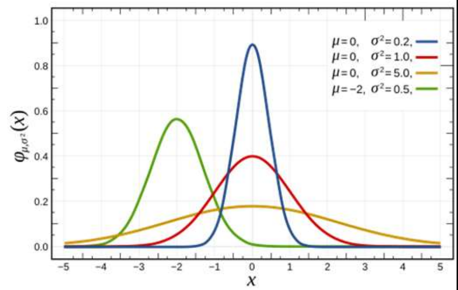
Normal Distribution

- A continuous random variable with parameters mean μ and variance σ^2
  - This pdf is symmetric about the mean μ and has the shape of the “bell curve”.
  - If μ=0 and σ=1, it is called the standard normal distribution



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

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Example: Pulse-rate of 50 students

Can calculate statistic such as sample mean (79.1) and sample standard deviation (7.6) but we can do more!

Pulse-rate (beats per minute) of 50 students									
89	68	92	74	76	65	77	83	75	87
85	64	79	77	96	80	70	85	80	80
82	81	86	71	90	87	71	72	62	78
77	90	83	81	73	80	78	81	81	75
82	88	79	79	94	82	66	78	74	72

Let’s plot these numbers

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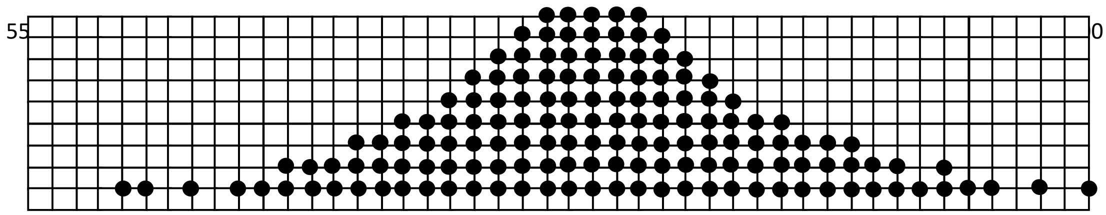
Distribution

The distribution approaches a normal symmetrical distribution

150 students



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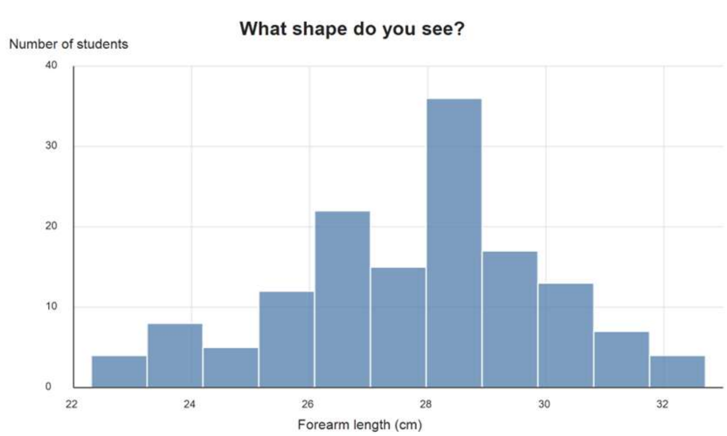
Self-reported forearm length



~140 students (male and female)

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Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Verifying validity of pdf
  - Consider Gaussian pdf with mean 0 and variance ½

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \qquad X \sim N(0,1)$$

$$\left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r\, dr\, d\theta = \pi$$

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Mean and MGF

$$E(X-\mu) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x-\mu) e^{-(x-\mu)^2/(2\sigma^2)} dx = 0$$

$$\sigma \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y e^{-y^2/2} dy = -\sigma \frac{1}{\sqrt{2\pi}}\ e^{-y^2/2}\Big|_{-\infty}^{\infty}$$

$$\Phi_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$$

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Variance

$$E[(X-\mu)^2] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \overset{\text{R.V.}}{(x-\mu)^2} \overset{\text{pdf}}{e^{-(x-\mu)^2/(2\sigma^2)}} dx = \sigma^2$$

$$\int u\,dv = uv - \int v\,du$$

$$\int ye^{-y^2/2}dy = -e^{-y^2/2}$$

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Properties

- Median = mean
  - Because of symmetry of the pdf about the mean
- Mode = mean
  - Set the first derivative of the pdf to 0

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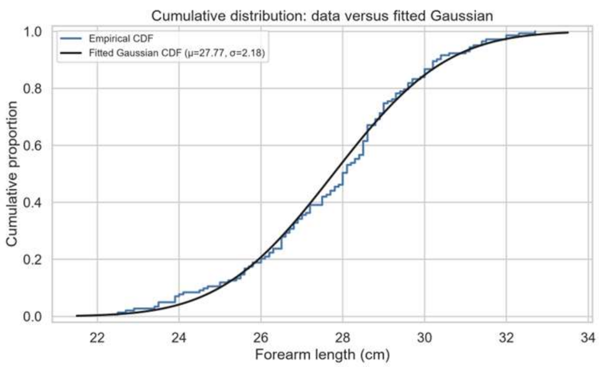
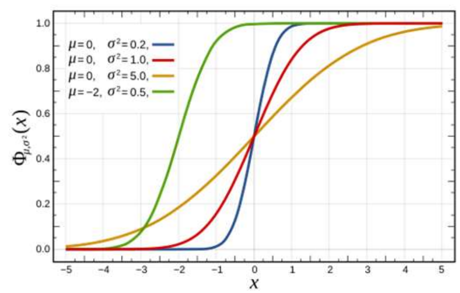
Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Cumulative distribution function





$$\Phi(x) = F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{x^2}{2}} dx$$

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